Preparing for IBM Domino backup and restore

Snap Creator Framework

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Preparing for IBM Domino backup and restore

Before you deploy the IBM Domino plug-in, make sure that your storage system and hosts meet minimum resource requirements. You also need to configure storage system layouts for databases, and optionally set up SnapMirror and SnapVault relationships.

For Snap Creator Server and Agent installation requirements, see the Snap Creator Framework 4.1.2 Installation Guide. Pay particular attention to the IBM Domino preinstallation requirements for the Agent host:

- On UNIX hosts, you must create symbolic links to IBM Domino shared object files.
- On Windows hosts, you must add the IBM Domino installation path to the PATH environment variable.

Storage layout requirements

A typical IBM Domino environment has at least three Domino volumes, one each for Domino data, Domino transaction logs, and the plug-in changeinfo directory. Many sites also have volumes for Domino DAOS and for view rebuilds.

The IBM Domino plug-in uses the changeinfo directory for changes recorded during backup operations and for copies of transaction logs used in up-to-the-minute restore operations. It is a best practice to store the changeinfo directory on a separate volume, to avoid inadvertently overwriting the information and to make it easier to back up.

You may also find it useful to have separate volumes for Domino DAOS (if it is enabled) and for view rebuilds. When Domino rebuilds a view (for example, when a user opens a view whose index has been deleted or when updall --R is run), it may generate temporary files to sort the data for rapid view rebuilding.

By default, these temporary files are located in the system's temporary folder or in the Domino data folder. IBM recommends changing the location of the temporary files to a different drive to distribute disk I/O and to ensure adequate space to rebuild views. To change the temporary folder used for view rebuilds, add the View_Rebuild_Dir setting to the notes.ini file.

The following table shows the preferred volume layout:

Volume	Contents	Notes
Volume 1	Domino data	FC, SAS, or SSD drives preferred.
Volume 2	Domino transaction logs	FC, SAS, or SSD drives preferred.
Volume 3	changeinfo	Stores changes recorded during backup operations and copies of transaction logs for use in up-to-the-minute restore operations.
Volume 4	View rebuild	Optional. Stores temp files created during index updates. Can use RAM disk. Add View_Rebuild_Dir setting to notes.ini file.

Volume	Contents	Notes
Volume 5	DAOS repository	Optional. Contains .dlo files from DAOS. Low I/O requirements make this a good candidate for SATA drives.

In virtual environments, guest-mounted disks are preferred.

SnapMirror and SnapVault setup

SnapMirror is disaster recovery technology, designed for failover from primary storage to secondary storage at a geographically remote site. SnapVault is archiving technology, designed for disk-to-disk Snapshot copy replication for standards compliance and other governance-related purposes.

Before you can use Snap Creator with these products, you need to configure a data-protection relationship between the source and destination volumes, then initialize the relationship.

The procedures in this section describe how to set up replication relationships in clustered Data ONTAP. You can find information about setting up these relationships in Data ONTAP operating in 7-Mode in the .

Preparing storage systems for SnapMirror replication

Before you can use to mirror Snapshot copies, you need to configure a data-protection relationship between the source and destination volumes, then initialize the relationship. Upon initialization, SnapMirror makes a Snapshot copy of the source volume, then transfers the copy and all the data blocks that it references to the destination volume. It also transfers any other, less recent Snapshot copies on the source volume to the destination volume.

- You must be a cluster administrator.
- For Snapshot copy verification on the destination volume, the source and destination Storage Virtual Machines (SVMs) must have a management LIF as well as a data LIF.

  The management LIF must have the same DNS name as the SVM. Set the management LIF role to data, the protocol to none, and the firewall policy to mgmt.

You can use the Data ONTAP command-line interface (CLI) or OnCommand System Manager to create a SnapMirror relationship. The following procedure documents CLI usage.

If you are storing database files and transaction logs on different volumes, you must create relationships between the source and destination volumes for the database files and between the source and destination volumes for the transaction logs.

The following illustration shows the procedure for initializing a SnapMirror relationship:

1. Identify the destination cluster.
2. On the destination cluster, use the volume create command with the -typeDP option to create a SnapMirror destination volume that is either the same or greater in size than the source volume.

   The language setting of the destination volume must match the language setting of the source volume.

   The following command creates a 2 GB destination volume named dstvolB in SVM2 on the aggregate node01_aggr:

   ```
   cluster2::> volume create -vserver SVM2 -volume dstvolB -aggregate
   node01_aggr -type DP
   -size 2GB
   ```

3. On the destination SVM, use the snapmirror create command with the -type DP parameter to create a SnapMirror relationship.

   The DP type defines the relationship as a SnapMirror relationship.

   The following command creates a SnapMirror relationship between the source volume srcvolA on SVM1 and the destination volume dstvolB on SVM2, and assigns the default SnapMirror policy DPDefault:

   ```
   SVM2::> snapmirror create -source-path SVM1:srcvolA -destination-path
   SVM2:dstvolB
   -type DP
   ```

   Do not define a mirror schedule for the SnapMirror relationship. does that for you when you create a backup schedule.

   If you do not want to use the default SnapMirror policy, you can invoke the snapmirror policy create command to define a SnapMirror policy.

4. Use the snapmirror initialize command to initialize the relationship.

   The initialization process performs a baseline transfer to the destination volume. SnapMirror makes a Snapshot copy of the source volume, then transfers the copy and all the data blocks it references to the destination volume. It also transfers any other Snapshot copies on the source volume to the destination volume.

   The following command initializes the relationship between the source volume srcvolA on SVM1 and the destination volume dstvolB on SVM2:

   ```
   SVM2::> snapmirror initialize -destination-path SVM2:dstvolB
   ```

Preparing storage systems for SnapVault replication

Before you can use to perform disk-to-disk backup replication, you need to configure a data-protection relationship between the source and destination volumes, then initialize the relationship. On initialization, SnapVault makes a Snapshot copy of the source volume, then transfers the copy and all the data blocks it references to the destination volume.

- You must be a cluster administrator.

You can use the Data ONTAP command-line interface (CLI) or OnCommand System Manager to create SnapVault relationships. The following procedure documents CLI usage.

> If you are storing database files and transaction logs on different volumes, you must create relationships between the source and destination volumes for the database files and between the source and destination volumes for the transaction logs.

The following illustration shows the procedure for initializing a SnapVault relationship:

1. Identify the destination cluster.
2. On the destination cluster, use the volume create command with the -typeDP option to create a SnapVault destination volume that is the same size as or larger than the source volume.

   > The language setting of the destination volume must match the language setting of the source volume.

   The following command creates a 2 GB destination volume named dstvolB in SVM2 on the aggregate node01_aggr:

   ```
   cluster2::> volume create -vserver SVM2 -volume dstvolB -aggregate
   node01_aggr -type DP
   -size 2GB
   ```

3. On the destination SVM, use the snapmirror policy create command to create a SnapVault policy.

   The following command creates the SVM-wide policy SVM1-vault:

   ```
   SVM2::> snapmirror policy create -vserver SVM2 -policy SVM1-vault
   ```

   > Do not define a cron schedule or Snapshot copy policy for the SnapVault relationship. does that for you when you create a backup schedule.

4. Use the snapmirror create command with the -type XDP parameter and the -policy parameter to create a SnapVault relationship and assign a vault policy.

   The XDP type defines the relationship as a SnapVault relationship.

The following command creates a SnapVault relationship between the source volume srcvolA on SVM1 and the destination volume dstvolB on SVM2, and assigns the policy SVM1-vault:

```
SVM2::> snapmirror create -source-path SVM1:srcvolA -destination-path
SVM2:dstvolB
-type XDP -policy SVM1-vault
```

5. Use the snapmirror initialize command to initialize the relationship.

   The initialization process performs a baseline transfer to the destination volume. SnapMirror makes a Snapshot copy of the source volume, then transfers the copy and all the data blocks it references to the destination volume.

   The following command initializes the relationship between the source volume srcvolA on SVM1 and the destination volume dstvolB on SVM2:

```
SVM2::> snapmirror initialize -destination-path SVM2:dstvolB
```

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